

H2BENCH · REQUIREMENTS

H2Bench requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2bench		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Decisions carried into the model (2026-10-02): R13 met at 24.4 kg; R9 at risk with the deoxidizer fitted; R14 USD 171 over the target
0.7	2026-10-02	Amish Chadha	Decisions of 2026-10-02: R13 counts the bench without its supply (10 mm deck decided); R9 deoxidizer decided; R8 interface proposal to H2Guard
0.6	2026-10-01	Amish Chadha	Constructable design (HBN-DDR-003); R13 mass and R15 guard clearance updated; R14 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from HBN-CAL-001; R14 scope states that H2Guard is costed in its own project (HBN-DDR-001)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with first-order status
0.1	2026-09-25	Amish Chadha	Initial scaffold

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These requirements were checked by calculation at TRL 3 in HBN-CAL-001 v0.3 (docs/04-calcs/01-sizing.md). Version 0.4 applies the decisions Amish accepted on 2026-09-25 (HBN-DDR-002): R14's limit is now the \$850 budget_usd (was \$450), with the bench power supply in the kit and H2Guard costed in its own project; R6 records the 325 kPa gauge relief and the 310 kPa gauge supply cut; and R10's target is restated from 300 kPa to 332 kPa, the relief setting plus the check valve, because a 300 kPa rating did not cover a relief event. Version 0.5 raises R14's limit to \$885, the budget Amish approved on 2026-09-26 to cover the priced BOM (HBN-DDR-002). Version 0.6 follows the constructable design of HBN-DDR-003 and reports R14 against budget_usd as a value-engineering target, not a limit (Amish, 2026-10-01). Eleven requirements are met on paper, R14 is USD 171 over its value-engineering target, three are at risk (R8, R9 and R10) and one (R16) cannot be verified until hardware exists. Version 0.8 carries the decisions of 2026-10-02 into the model: R13 becomes met (24.4 kg for the bench as moved) and R9 moves from not met to at risk (deoxidizer fitted, no supplier data). "Met" means met by calculation or design review, not by test.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status (HBN-CAL-001)	Verification (TRL 3 or later)
R1	Measure the efficiency of each stage	Electrolyzer, storage, fuel cell and round trip, each to within ± 5 % relative	Met: hydrogen per fill ± 1.5 %, worst stage ± 1.8 %	Uncertainty calculation (HBN-CAL-001 section 6)
R2	Complete a full cycle in one lesson	Fill and discharge within 60 min of run time	Met: 17.7 min fill plus 29.0 min discharge, 46.7 min	Energy and flow calculation
R3	Electrolyzer power	100 W or less electrical input	Met: 70.2 W (79 W with aged cells)	Calculation; stack datasheet later
R4	Fuel cell power	50 W or less; 10 W or more net to run a visible load	Met: 10.8 W net, a 0.8 W margin (9.8 W if cells fall to 0.55 V)	Calculation; stack datasheet later
R5	Limit the hydrogen inventory	10 L or less at 20 °C and 101.3 kPa	Met: 8.5 L at 300 kPa gauge, 9.0 L at the 325 kPa gauge relief setting, 9.9 L at full relief lift on a 15 °C day	Gas law calculation
R6	Limit storage pressure	Working pressure 300 kPa gauge or less; electrolyzer supply cut at 310 kPa gauge; relief at 350 kPa gauge or less; vessel rated 1 MPa or more	Met by design: cut 310 kPa gauge, relief 325 kPa gauge; hoop stress 18 MPa at 1 MPa	Component ratings
R7	No flammable mixture in the room from a total release	Full inventory mixed into a 30 m ³ room below 25 % of the lower flammable limit	Met: 0.030 % by volume, 0.75 % of LFL	Calculation
R8	Interlock shuts down on a leak	H2Guard alarm cuts electrolyzer power and closes the tank solenoid within 2 s; fan runs	At risk: 1.1 s from H2Guard's TRL 2 figures, but H2Guard is at TRL 2 and its valve is 24 V against H2Bench's 12 V solenoid. Decided 2026-10-02 to propose to H2Guard a 24 V normally closed tank solenoid powered from H2Guard and its alarm contact in series with the 310 kPa cut relay	H2Guard design review
R9	Hydrogen quality at the fuel cell	Purity as the stack requires (the 12 W class specifies 99.995 % or better)	At risk: water is removed by the drier and a catalytic deoxidizer (BOM line 19) now sits between separator and drier. At an assumed 99 % conversion the residual oxygen is 10 to 200 ppm for 0.1 to 2 % crossover, against 50 ppm for all impurities, and there is no supplier data. The deoxidizer is removable only if the stack supplier's data show oxygen	Supplier data, later gas test

ID	Requirement	Target	Status (HBN-CAL-001)	Verification (TRL 3 or later)
			within the 99.995 % limit at the lowest operating current	
R10	Electrolyzer tolerates tank back-pressure	H2 side rated to 332 kPa or more over the O2 side (relief setting plus check valve; was 300 kPa)	At risk: the stack must hold 307 kPa while filling, 317 kPa at the supply cut and 332 kPa at the relief setting; no rating in hand	Supplier data
R11	Water quality	Conductivity 1 μ S/cm or less at the stack inlet, shown to students	Met by design (mixed-bed resin and conductivity check)	Datasheet
R12	Log data students can analyze	Voltage, current, power, tank pressure and temperature at 1 Hz, saved as CSV	Met by design: 2,804 rows, about 224 kB per cycle	Firmware sketch review
R13	Fit an existing lab bench	Footprint 1,000 x 500 mm or less; height 800 mm or less; mass 25 kg or less for the bench as moved (the bench supply ships in the kit but sits beside the bench, HBN-DDR-003 A1, 2026-10-02)	Met: 900 x 450 x 753 mm, 24.4 kg for the bench as moved with an aluminium tank (the supply stands beside it; 27.4 kg with the supply); the 10 mm deck of 2026-10-02 is in the model	Parametric model and mass estimate
R14	Low cost and buildable	Parts cost reported against the USD 885 value-engineering target (budget_usd , was \$450, then \$850), including the bench power supply and excluding H2Guard, which is costed in its own project and shipped with the bench; hand tools only; no welding	USD 1056, USD 171 over the value-engineering target (USD 991 without the bench supply); hand tools only, no welding (HBN-DDR-003)	Priced BOM
R15	Guard people from pressure parts	Tank inside a guard; relief and vent piped to the hood; no pressurized fitting that can be opened without a tool	Met by design: 50 mm guard clearance; relief and vent lines to the outlet hanger under the canopy	Design review
R16	Set up and shut down quickly	Setup 15 min or less; tank vented and bench safe in 5 min or less	Not verifiable at TRL 3: the vent takes 3.0 min at the needle valve setting; setup needs a walk-through with hardware	Walk-through when hardware exists

■ Assumptions

- Electrolyzer: 4 cells at about 1.95 V and 9 A, Faraday efficiency 95 %; hydrogen higher heating value (HHV) 285.8 kJ/mol.
- Tank: 2.000 L internal plus 0.15 L of lines at tank pressure, filled from 70 to 300 kPa gauge (171 to 401 kPa absolute) at 20 °C; below 70 kPa gauge the regulator cannot hold the fuel cell supply. The full assumption list is in HBN-CAL-001 Table 1.
- Fuel cell: 13 cells at 0.60 V and 1.5 A, fuel utilization 95 % (the rest purged), about 0.9 W for fan and controls.

- Room: 30 m³, well mixed. The canopy hood exists because mixing near the leak is not instant; see HBN-PRC-001.
- Round-trip figures use the HHV basis. On a lower heating value (LHV) basis the hydrogen energy is about 15 % lower and the fuel cell efficiency correspondingly higher.